

## Question

**X** and **Y** are elements in the same group and have similar chemical properties. The dioxide of **X**, **A**, has widespread applications in organic chemistry, and its aqueous solution can be reduced by sulfur dioxide to generate the element **X** (reaction 1); **A** can react with **H** to obtain ion **B** and ion **C** (reaction 2), **B** has one more **X** atom than **C** and the charge remains unchanged. The chloride of **X**, **D**, reacts with  $\text{NH}_3$  to produce an explosive substance **E** (reaction 3). **D** reacts with **A** to obtain **F**, and **F** reacts with Cu element, and the products include **A** and **G** (reaction 4). **F** can also react with excess sulfur element (reaction 5), and the products include **A** and **G**, as well as **I**, which has a similar structure to **G**.  $\text{Z}^{2-}$  containing 57.01% **Y** reacts with sulfite to generate the element **Y** and the oxygen-containing anion **H** (reaction 6). The sodium salt of **H** is a common reagent in analytical chemistry and reacts with  $\text{I}_2$  to convert to **C**.

Which of the following statements are correct?

- B** contains 11 atoms
- Molecule **E** contains a 4-fold rotation axis
- The ratio of reaction coefficients of each reactant and product in reaction 3 is 12:64:3:4:48
- The ratio of reaction coefficients of each reactant and product in reaction 5 is 6:4:3:2:1
- Sulfide ions are also produced in reaction 6
- I** is a reddish-brown gas
- Reaction 2 does not require acid participation
- Reaction 4 also produces a water-insoluble white precipitate.

**A.** All other options are incorrect

**B.** 2.3.6.

**C.** 1.2.3.4.6.

**D.** 1.4.5.

**E.** 2.3.5.

**F.** 3.6.8.

**G.** 2.4.7.8.

**H.** 1.5.8.

**I.** 2.4.7.

**J.** 2.3.

**K.** 4.5

**L.** 2.8.

## Answer

Correct Answer: **D**

## Detailed Explanation

According to the knowledge of elemental chemistry, **X** is *Se*, **Y** is *Te*

### CHECKPOINT

1 PTS

**X** is *Se*, **Y** is *Te*

**A** is the dioxide of *Se*, so **A** is  $SeO_2$ ; from the mass fraction, we can know that  $Z^{2-}$  should be  $TeS_3^{2-}$ . From the role of the sodium salt of **H** in analytical chemistry, we can know that **H** is  $S_2O_3^{2-}$ . Thus we can deduce that **C** is  $S_4O_6^{2-}$ , and from the fact that **B** has one more *Se* atom than **C**, we can deduce that the chemical formula of **B** is  $SeS_4O_6^{2-}$ .

### CHECKPOINT

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**A** is  $SeO_2$

### CHECKPOINT

1 PTS

$Z^{2-}$  is  $TeS_3^{2-}$

CHECKPOINT

1 PTS

**H** is  $S_2O_3^{2-}$

CHECKPOINT

1 PTS

**C** is  $S_4O_6^{2-}$

CHECKPOINT

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**B** is  $SeS_4O_6^{2-}$

The chloride of *Se*, **D**, should be  $SeCl_4$ , which corresponds to **E** when reacted with ammonia to yield  $Se_4N_4$ .

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**D** should be  $SeCl_4$

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**E** is  $Se_4N_4$

**D** reacts with **A** to yield **F**, which should be  $SeOCl_2$ , and can react with Cu to yield **G** as  $Se_2Cl_2$ , and reacts with sulfur to yield **I** with a similar structure:  $S_2Cl_2$ .

## CHECKPOINT

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**F** is  $SeOCl_2$

## CHECKPOINT

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**G** is  $Se_2Cl_2$

## CHECKPOINT

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**I** is  $S_2Cl_2$

**B** contains  $4+6+1=11$  atoms, 1 is correct.

The molecule **E** contains a 4-fold rotoinversion axis but not a rotation axis, 2 is incorrect.

**I** is a red-brown liquid, 6 is incorrect.

Reaction 1.  $SeO_2 + 2SO_2 + 2H_2O = 2H_2SO_4 + Se$

## CHECKPOINT

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Reaction 1.  $SeO_2 + 2SO_2 + 2H_2O = 2H_2SO_4 + Se$

Reaction 2.  $4H^+ + SeO_2 + 4S_2O_3^{2-} = S_4O_6^{2-} + SeS_4O_6^{2-} + 2H_2O$

Reaction 2 requires acid participation, 7 is incorrect.

## CHECKPOINT

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Reaction 2.  $4H^+ + SeO_2 + 4S_2O_3^{2-} = S_4O_6^{2-} + SeS_4O_6^{2-} + 2H_2O$

Reaction 3.  $12SeCl_4 + 64NH_3 = 3Se_4N_4 + 2N_2 + 48NH_4Cl$

The ratio of the reaction coefficients of each reactant and product in reaction 3 is 12:64:3:2:48, 3 is incorrect.

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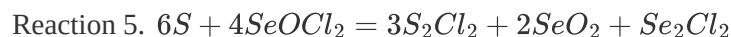
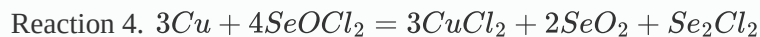
Reaction 3.  $12SeCl_4 + 64NH_3 = 3Se_4N_4 + 2N_2 + 48NH_4Cl$

Reaction 4.  $3Cu + 4SeOCl_2 = 3CuCl_2 + 2SeO_2 + Se_2Cl_2$

$CuCl_2$  is soluble in water, 8 is incorrect.

## CHECKPOINT

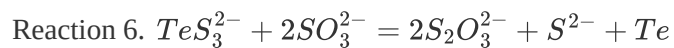
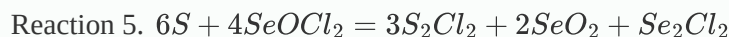
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The ratio of the reaction coefficients of each reactant and product in reaction 5 is 6:4:3:2:1, 4 is correct.

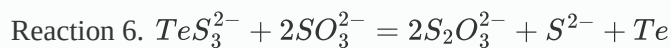
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## CHECKPOINT

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Sulfide ions are also generated in reaction 6, 5 is correct.

It can be found that only 1, 4, and 5 are correct, choose D.